

COURSE DESCRIPTION

Course Number	0418-220	Course Title	Programming in C and UNIX
Credit Hours	3 (3:1:3)	Course Coordinator	Dr. Naser AlDuaij
Compulsory	Yes	Prerequisite	0418-141

Description

This course is designed for students with prior programming experience to learn programming in C in a Unix-like programming environment. Features of the C language that are covered include arrays, structures and unions, file processing, pointers, pointer arithmetic, casting and dynamic memory allocation (malloc, calloc and free). Data structures that are emphasized include lists, queues, stacks, and hash tables. Students will learn to use tools provided in UNIX such as emacs/vi, make, and gdb to assist them in the design, testing and debugging of their programs. Students will learn about regular expressions and grep and will be able to use a scripting language such as Perl/Python to solve simple problems.

Textbook

Title: System Programming with C and Unix
Author: Adam Hoover
Publisher: Pearson, latest edition

Reference

Title: C Programming Language
Author: Brian W. Kernighan, Dennis M. Ritchie
Publisher: Prentice-Hall; 2nd edition

Major Topics Covered in the Course

	Topic	Hours
1	Overview of the UNIX operating system and environment, basic Unix commands and Unix editor (vi/vim), VMs, ssh.	3
2	C programming: Standard I/O and formatted I/O. GDB debugger.	3
3	C programming using struct, union, enumeration, typedef, array of pointers, C strings, and pointer to function. C preprocessor.	6
4	Bit-fields manipulation, command-line arguments, redirect input/output and makefile.	4
5	Standard streams and files (random and sequential access).	3
6	Dynamic memory allocation: malloc, calloc, realloc and free. Linked list, stack, queue, tree and hash table. Valgrind	5
7	Unix sessions, environments, files and directories.	2
8	Unix Shell Script Programming: Bash	5
9	Regular expression using grep, awk, and sed.	5
10	Perl or Python programming	3
11	Exams	3
	Total	42

Labs

Topics	Weeks
Basics of UNIX/Linux: vi editor, UNIX commands	1
GDB debugger	2
GDB debugger	3
Complex C data type: pointer, structure and union	4
Redirect input/output and makefile	5
C Files (binary, text, random and sequential)	6
Dynamic memory allocation 1 & Valgrind	7
Dynamic memory allocation 2 & Valgrind	8
Files and directories.	9
Shell Script	10
Shell Script	11
Shell Script and grep	12
awk and sed.	13
Perl or Python programming.	14

Grading

Midterm 1 and 2	40%
Labs & HW	20%
Final Exam	40%

Course Outcomes

1. Understand and explain the basic tools and interfaces of UNIX-based systems.
2. Design & implement efficient C programs under the UNIX/Linux environment.
3. Design & implement Bash scripts.
4. Use debugging tools.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	L	L				
2	L	H				
3	L	H				
4	L	H				

Last updated Spring 2023